Linxup Push API

Contents

Overview

This document describes how we push data to customers in real time. Customers are required to provide the following for this API to be configured:

- One or more of the following SSL secured URL end points that we can send JSON to:
  - SSL secured URL for POSTing GPS positions and fence events.
  - SSL secured URL for POSTing stops.
  - SSL secured URL for POSTing trips.
  - SSL secured URL for POSTing usage hour segments.
  - SSL secured URL for POSTing alerts.
- Bearer token that will be included in every call to the customer's URLs in the "Authentication" header.
- A single URL can be provided for all data.
- We plan to add items to the pushed data in the future, so please be flexible in your JSON parsing.

Positions

Whenever a GPS position is received from a device, it will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Data sent

- pushType – should be set to “POSITION”.
- deviceNumber - unique IMEI of the device.
- deviceSerialNumber – manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.
- deviceTypeDescription - “Linxup AT3", for example.
- driverNumber – unique alphanumeric value assigned to this device by customer.
- firstName – first name assigned to this device by customer.
- lastName – last name assigned to this device by customer.
- fenceName - name of geofence which contains the device.
- geofenceUUID - unique identifier for geofence which contains the device.
- fleetName – name of the group device belongs to (set to “Default” if no groups set up)
- vin – VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make – make of vehicle device is installed in (e.g. “Chevrolet”).
- model – model of vehicle device is installed in (e.g. “Silverado”).
- year – year of vehicle device is installed in (e.g. “2011”).
- date - milliseconds since epoch.
- formattedDate - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- latitude
- longitude
- altitude – in meters.
- speed - in mph.
- heading - the heading (e.g. "N", "SE", etc.). May be "None" if not moving.
- direction - degrees from north. See Appendix A: Direction.
- odo – odometer in miles (some devices provide actual odo, others virtual calculated via GPS).
- battery – voltage of battery. In case of asset trackers 2 values separated by “/” are provided – external and internal voltages (e.g. “12.1” or “14.1/4.0”). For asset trackers, internal voltage of 4.2V means fully charged while less than 3.3 means discharged.
- fuelLevel - only available on newer plugins. Reported as a percentage.
- estSpeedLimit - estimated speed limit at the lat/lon. Not guaranteed to be present.
- speeding - whether the device was considered speeding (above the estimated speed limit). If the speed is 0, may be null.
- behaviorCd - an alert for the position.
    - "HAC" - rapid acceleration.
    - "HBR" - harsh braking.
- street – generally not present.
- city - generally not present.

- stateCode - generally not present.
- postalCode - generally not present.
- countryCode - generally not present.
- currentState – STOPPED or MOVING are the two possible values.
- engineOn – true or false.
- sensorData - array of sensor readings each of which has:
    - sensorNumber - number of sensor (starts at 0)
    - temperature - temperature in degrees Fahrenheit
    - humidity - relative humidity percentage
- customerUUID - unique identifier for customer.

Example


```
{
    "altitude": 174,
    "battery": "12.4",
    "behaviorCd": null,
    "city": null,
    "countryCode": null,
    "currentState": "STOPPED",
    "customerUUID”: "6499d7a4-529b-45b9-a8b9-910dc22fc655",
    "date": 1442392916000,
    "deviceNumber": "351112223334444",
    "deviceSerialNumber": "SN123456",
    "deviceTypeDescription": "Linxup Plugin",
    "deviceUUID": "366a471b-3064-4cef-8f20-b6c4eed33687",
    "direction": 0,
    "driverNumber": "DRV01234",
    "engineOn": false,
    "estSpeedLimit": null,
    "fenceName": "Depot 1",
    "geofenceUUID" : "0efc215a-47d7-4b07-89fc-89f8fb3b5906",
    "firstName": "Bill",
    "fleetName": "Default",
    "formattedDate": "2015-09-16T03:41:56-0500",
    "fuelLevel": "88.1%",
    "heading": "N",
    "lastName": "McCoy",
    "latitude": 38.67386,
    "longitude": -90.5117,
    "make": "Chevrolet",
    "model": "Colorado",
    "odo": 15124,
    "postalCode": null,
    "pushType": "POSITION",
    "sensorData": [
        {
            "humidity": 29.71,
            "sensorNumber": 0,
            "temperature": 24.91
        }
    ],
    "speed": 0,
    "speeding": false,
```

```
    "stateCode": null,
    "street": null,
    "vin": "1WAHSBE44G3230973",
    "year": "2013"
}
```

Fence Events

Whenever a GPS position is received from a device that results in a fence being entered or exited, it will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Data sent

- pushType – should be set to "FENCE_EVENT".
- deviceNumber - unique IMEI of the device.
- deviceSerialNumber – manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.
- deviceTypeDescription - "Linxup AT3", for example.
- driverNumber – unique alphanumeric value assigned to this device by customer.
- firstName – first name assigned to this device by customer.
- lastName – last name assigned to this device by customer.
- fenceName - name of geofence which contains the device.
- geofenceUUID - unique identifier for geofence which contains the device.
- fleetName – name of the group device belongs to (set to "Default" if no groups set up)
- vin – VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make – make of vehicle device is installed in (e.g. "Chevrolet").
- model – model of vehicle device is installed in (e.g. "Silverado").
- year – year of vehicle device is installed in (e.g. "2011").
- fenceEventCd - FENCE_ENTER or FENCE_EXIT.
- date - milliseconds since epoch.
- formattedDate - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- latitude
- longitude
- altitude – in meters.
- speed - in mph.
- heading - the heading (e.g. "N", "SE", etc.). May be "None" if not moving.
- direction - degrees from north. See Appendix A: Direction.
- odo – odometer value in miles (some devices provide actual odo, others virtual calculated via GPS).
- battery – voltage of battery. In case of asset trackers 2 values separated by "/" are provided – external and internal voltages (e.g. "12.1" or "14.1/4.0"). For asset trackers, internal voltage of 4.2V means fully charged while less than 3.3 means discharged.
- fuelLevel - only available on newer plugins. Reported as a percentage.
- estSpeedLimit - estimated speed limit at the lat/lon. Not guaranteed to be present.
- speeding - whether the device was considered speeding (above the estimated speed limit). If the speed is 0, may be null.
- behaviorCd - an alert for the position.
    - "HAC" - rapid acceleration.
    - "HBR" - harsh braking.
- street – generally not present.

- city - generally not present.
- stateCode - generally not present.
- postalCode - generally not present.
- countryCode - generally not present.
- currentState – STOPPED or MOVING are the two possible values.
- engineOn – true or false
- customerUUID - unique identifier for customer.

Example


```
{
    "altitude": 174,
    "battery": "12.4",
    "behaviorCd": null,
    "city": null,
    "countryCode": null,
    "currentState": "STOPPED",
    "customerUUID”: "6499d7a4-529b-45b9-a8b9-910dc22fc655",
    "date": 1442392916000,
    "deviceNumber": "351112223334444",
    "deviceSerialNumber": "SN123456",
    "deviceTypeDescription": "Linxup Plugin",
    "deviceUUID": "366a471b-3064-4cef-8f20-b6c4eed33687",
    "direction": 0,
    "driverNumber": "DRV01234",
    "engineOn": false,
    "estSpeedLimit": null,
    "fenceEventCd": "FENCE_ENTER",
    "fenceName": "Depot 1",
    "firstName": "Bill",
    "fleetName": "Default",
    "formattedDate": "2015-09-16T03:41:56-0500",
    "fuelLevel": "88.1%",
    "geofenceUUID": "0efc215a-47d7-4b07-89fc-89f8fb3b5906",
    "heading": "N",
    "lastName": "McCoy",
    "latitude": 38.67386,
    "longitude": -90.5117,
    "make": "Chevrolet",
    "model": "Colorado",
    "odo": 15124,
    "postalCode": null,
    "pushType": "FENCE_EVENT",
    "speed": 0,
    "speeding": false,
    "stateCode": null,
    "street": null,
    "vin": "1WAHSBE44G3230973",
    "year": "2013"
}
```

Stops

Whenever a stop concludes, it will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Data sent

- pushType – should be set to “STOP”.
- deviceNumber - unique IMEI of the device.
- deviceSerialNumber – manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.

- deviceTypeDescription - "Linxup AT3", for example.
- driverNumber – unique alphanumeric value assigned to this device by customer.
- firstName – first name assigned to this device by customer.
- lastName – last name assigned to this device by customer.
- fenceName - name of geofence which contains the device.
- geofenceUUID - unique identifier for geofence which contains the device.
- fleetName – name of the group device belongs to (set to “Default” if no groups set up)
- vin – VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make – make of vehicle device is installed in (e.g. “Chevrolet”).
- model – model of vehicle device is installed in (e.g. “Silverado”).
- year – year of vehicle device is installed in (e.g. “2011”).
- latitude
- longitude
- startDateTime - milliseconds since epoch.
- formattedStartDateTime - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- endDateTime - milliseconds since epoch.
- formattedEndDateTime - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- durationMinutes – the duration of the stop in minutes.
- street – street address of the stop (not guaranteed to be present).
- city – city of the stop (not guaranteed to be present).
- stateCode – state of the stop (not guaranteed to be present).
- postalCode – postal code of the stop (not guaranteed to be present).
- countryCode – country code of the stop (not guaranteed to be present).
- stopType – “Engine Off” or “Idling”
- customerUUID - unique identifier for customer.

Example

```
{
    "city": "Weldon Spring",
    "countryCode": "USA",
    "customerUUID”: "6499d7a4-529b-45b9-a8b9-910dc22fc655",
    "deviceNumber": "351112223334444",
    "deviceSerialNumber": "SN123456",
    "deviceTypeDescription": "Linxup Plugin",
```

```
    "deviceUUID": "366a471b-3064-4cef-8f20-b6c4eed33687",
    "driverNumber": "DRV01234",
    "durationMinutes": 60,
    "endDateTime": 1506436325574,
    "fenceName": "Depot 1",
    "firstName": "Bill",
    "fleetName": "Default",
    "formattedEndDateTime": "2017-09-26T09:32:05-0500",
    "formattedStartDateTime": "2017-09-26T08:32:05-0500",
    "geofenceUUID": "0efc215a-47d7-4b07-89fc-89f8fb3b5906",
    "lastName": "McCoy",
    "latitude": 38.7156,
    "longitude": -90.6702,
    "make": "Chevrolet",
    "model": "Colorado",
    "postalCode": "63304",
    "pushType": "STOP",
    "startDateTime": 1506432725574,
    "stateCode": "MO",
    "stopType": "Engine Off",
    "street": "612 Northbrook Court",
    "vin": "1WAHSBE44G3230973",
    "year": "2013"
}
```

Usage Hours

Whenever an asset concludes a usage session (ignition off received from a properly wired device), it will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Data sent

- pushType – should be set to “USAGE_HOURS”.
- deviceNumber - unique IMEI of the device.
- deviceSerialNumber – manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.

- deviceTypeDescription - "Linxup AT3", for example.
- driverNumber – unique alphanumeric value assigned to this device by customer.
- firstName – first name assigned to this device by customer.
- lastName – last name assigned to this device by customer.
- fenceName - name of geofence which contains the device.
- geofenceUUID - unique identifier for geofence which contains the device.
- fleetName – name of the group device belongs to (set to “Default” if no groups set up)
- vin – VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make – make of vehicle device is installed in (e.g. “Chevrolet”).
- model – model of vehicle device is installed in (e.g. “Silverado”).
- year – year of vehicle device is installed in (e.g. “2011”).
- startDateTime - milliseconds since epoch.
- formattedStartDateTime - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- startLatitude
- startLongitude
- startAddress - if available.
- startFenceName - if available.
- endDateTime - milliseconds since epoch.
- formattedEndDateTime - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- endLatitude
- endLongitude
- endAddress - if available.
- endFenceName - if available.
- durationMinutes - duration in minutes.
- customerUUID - unique identifier for customer.

Example


```
{
    "customerUUID”: "6499d7a4-529b-45b9-a8b9-910dc22fc655",
    "deviceNumber": "351112223334444",
    "deviceSerialNumber": "SN123456",
    "deviceTypeDescription": "Linxup Plugin",
    "deviceUUID": "366a471b-3064-4cef-8f20-b6c4eed33687",
    "driverNumber": "DRV01234",
```

```
    "durationMinutes": 60,
    "endAddress": "123 Circle Dr, Chesterfield, MO 63017",
    "endDateTime": 1506436325574,
    "endFenceName": "Depot 1",
    "endLatitude": 38.7156,
    "endLongitude": -90.6702,
    "fenceName": "Depot 1",
    "firstName": "Bill",
    "fleetName": "Default",
    "formattedEndDateTime": "2017-09-26T09:32:05-0500",
    "formattedStartDateTime": "2017-09-26T08:32:05-0500",
    "geofenceUUID": "0efc215a-47d7-4b07-89fc-89f8fb3b5906",
    "lastName": "McCoy",
    "make": "Chevrolet",
    "model": "Colorado",
    "pushType": "USAGE_HOURS",
    "startAddress": "123 Circle Dr, Chesterfield, MO 63017",
    "startDateTime": 1506432725574,
    "startFenceName": "Depot 1",
    "startLatitude": 38.7156,
    "startLongitude": -90.6702,
    "vin": "1WAHSBE44G3230973",
    "year": "2013"
}
```

Trips

Whenever a vehicle concludes a usage session (ignition off received from a properly wired device), it will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Data sent

- pushType – should be set to "TRIP".
- deviceNumber - unique IMEI of the device.
- deviceSerialNumber – manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.

- deviceTypeDescription - "Linxup AT3", for example.
- driverNumber – unique alphanumeric value assigned to this device by customer.
- firstName – first name assigned to this device by customer.
- lastName – last name assigned to this device by customer.
- fleetName – name of the group device belongs to (set to "Default" if no groups set up)
- vin – VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make – make of vehicle device is installed in (e.g. "Chevrolet").
- model – model of vehicle device is installed in (e.g. "Silverado").
- year – year of vehicle device is installed in (e.g. "2011").
- startDateTime - milliseconds since epoch.
- formattedStartDateTime - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- startLatitude
- startLongitude
- startAddress - if available.
- endDateTime - milliseconds since epoch.
- formattedEndDateTime - ISO 8601 format (yyyy-MM-dd'T'HH:mm:ssZ).
- endLatitude
- endLongitude
- endAddress - if available.
- durationMinutes - duration in minutes.
- distanceMiles – distance in miles (rounded to nearest mile).
- authorizedMiles – distance driven within allowed hours (rounded to nearest mile).
- unauthorizedMiles – distance driven outside of allowed hours (rounded to nearest mile).
- distanceMilesDetailed – distance in miles (to 3 decimal places).
- authorizedMilesDetailed – distance driven within allowed hours (to 3 decimal places).
- unauthorizedMilesDetailed – distance driven outside of allowed hours (to 3 decimal places).
- customerUUID - unique identifier for customer.

Example


```
{
    "authorizedMiles": 25,
    "authorizedMilesDetailed": 24.781,
    "customerUUID": "6499d7a4-529b-45b9-a8b9-910dc22fc655",
    "deviceNumber": "351112223334444",
```

```
    "deviceSerialNumber": "SN123456",
    "deviceTypeDescription": "Linxup Plugin",
    "deviceUUID": "366a471b-3064-4cef-8f20-b6c4eed33687",
    "distanceMiles": 25,
    "distanceMilesDetailed": 24.781,
    "driverNumber": "DRV01234",
    "durationMinutes": 60,
    "endAddress": "123 Circle Dr, Chesterfield, MO 63017",
    "endDateTime": 1506436325574,
    "endLatitude": 38.7156,
    "endLongitude": -90.6702,
    "firstName": "Bill",
    "fleetName": "Default",
    "formattedEndDateTime": "2017-09-26T09:32:05-0500",
    "formattedStartDateTime": "2017-09-26T08:32:05-0500",
    "lastName": "McCoy",
    "make": "Chevrolet",
    "model": "Colorado",
    "pushType": "TRIP",
    "startAddress": "123 Circle Dr, Chesterfield, MO 63017",
    "startDateTime": 1506432725574,
    "startLatitude": 38.7156,
    "startLongitude": -90.6702,
    "unauthorizedMiles": 0,
    "unauthorizedMilesDetailed": 0,
    "vin": "1WAHSBE44G3230973",
    "year": "2013"
}
```

Alerts

Whenever a vehicle generates an alert, it will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Alert types are:

- `FIRST_IGNITION_OF_DAY`
- `GEOFENCE_ENTERED`
- `GEOFENCE_EXITED`
- `HARSH_BRAKING`
- `HIGH_SPEED – Vehicle has exceed a hard upper limit set by the customer`
- `SPEEDING - Vehicle has exceed the posted speed limit (including buffer)`
- `IDLE_START – Vehicle has started idling (engine on, not moving)`
- `IDLE_END`
- `NO_SIGNAL – Device has not reported in (12 hours for vehicles, 50 for assets)`
- `RAPID_ACCELERATION`
- `SENSOR_USE_EXCEEDS_LIMIT – Sensor has gone HIGH more than allowed limit`
- `SENSOR_UNAUTHORIZED_USE – Sensor has gone HIGH outside of authorized hours`
- `POWER_ON – Device has powered on (this is also known as TAMPER)`
- `UNAUTHORIZED_USE – Vehicle is being used outside of authorized hours`
- `ASSET_LOW_BATTERY`
- `DIAGNOSTIC_TROUBLE_CODE_NEW – Vehicle has thrown a new diagnostic code`
- `DIAGNOSTIC_TROUBLE_CODE_CLEAR – Vehicle has cleared a diagnostic code`
- `SENSOR_ONE_HIGH`
- `SENSOR_ONE_LOW`
- `SENSOR_TWO_HIGH`
- `SENSOR_TWO_LOW`

Note that all alert types are not available on all devices. For example, ASSET_LOW_BATTERY is not available on OBD2 plugs. Some, like SENSOR types, require additional wiring. Some, like DIAGNOSTIC types, are only available on OBD2 plugs.

Data sent

- pushType – should be set to “ALERT”.
- deviceNumber - unique IMEI of the device.
- deviceSerialNumber – manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.

- deviceTypeDescription - "Linxup AT3", for example.
- driverNumber – unique alphanumeric value assigned to this device by customer.
- firstName – first name assigned to this device by customer.
- lastName – last name assigned to this device by customer.
- fenceName - name of geofence which contains the device.
- geofenceUUID - unique identifier for geofence which contains the device.
- fleetName – name of the group device belongs to (set to “Default” if no groups set up)
- vin – VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make – make of vehicle device is installed in (e.g. “Chevrolet”).

- model – model of vehicle device is installed in (e.g. “Silverado”).
- year – year of vehicle device is installed in (e.g. “2011”).
- alertCode – The type of alert.
- alertShortDesc – Short description, suitable for a subject line.
- alertDesc – Long description of the alert.
- alertDateTime - milliseconds since epoch.
- formattedAlertDateTime - ISO 8601 format (yyyy-MM-dd’T'HH:mm:ssZ).
- customerUUID - unique identifier for customer.

Example


```
{
    "alertCode": "NO_SIGNAL",
    "alertDateTime": 1506432725574,
    "alertDesc": "Tracking device has not reported a location for more than 12 hours.",
    "alertShortDesc": "No Signal Alert",
    "customerUUID”: "6499d7a4-529b-45b9-a8b9-910dc22fc655",
    "deviceNumber": "351112223334444",
    "deviceSerialNumber": "SN123456",
    "deviceTypeDescription": "Plugin",
    "deviceUUID": "366a471b-3064-4cef-8f20-b6c4eed33687",
    "driverNumber": "DRV01234",
    "fenceName": "Depot 1",
    "firstName": "Bill",
    "fleetName": "Default",
    "formattedAlertDateTime": "2017-09-26T08:32:05-0500",
    "geofenceUUID": "0efc215a-47d7-4b07-89fc-89f8fb3b5906",
    "lastName": "McCoy",
    "make": "Chevrolet",
    "model": "Colorado",
    "pushType": "ALERT",
    "vin": "1WAHSBE44G3230973",
    "year": "2013"
}
```

Device Updates

Whenever information about a tracker or the vehicle in which it's installed are updated, a message will be posted to the provided URL in JSON format. HTTP status code of 201 should be returned in response.

Data sent

- deviceNumber - unique IMEI of the device.
- deviceSerialNumber - manufacturer serial number of the device.
- deviceUUID - unique identifier for the device.
- driverNumber - unique alphanumeric value assigned to this device by customer.
- customerUUID - unique identifier for customer.
- firstName - first name assigned to this device by customer.
- lastName - last name assigned to this device by customer.
- deviceTypeDescription - "Linxup AT3", for example.
- vin - VIN of vehicle device is installed in. Automatically obtained on most newer plugins.
- make - make of vehicle device is installed in (e.g. "Chevrolet").
- model - model of vehicle device is installed in (e.g. "Silverado").
- year - year of vehicle device is installed in (e.g. "2011").
- fleetName - The name of the group that the device is assigned to.

Example


```
{
 "deviceNumber": '3519340418424',
 "deviceSerialNumber": 'VB581556027',
 "deviceUUID": '43b1b40e-096d-4660-8c3c-a46bc98c72ee',
 "driverNumber": null,
 "customerUUID": 'db20426d-d502-4e43-ab32-f451153c0a77',
 "firstName": 'Example',
 "lastName": 'Device',
 "deviceTypeDescription": 'Linxup Wired (3G Dual-Band 2160)',
 "vin": "1WAHSBE44G3230973",
 "make": "Chevrolet",
 "model": "Colorado
 "year": "2013"
 "fleetName": 'Fake No Group'
}
```

Appendix A

Direction

Whenever direction in degrees from north is provided, you can interpret it as follows:

- 0-22 or 338-360 = N
- 23-67 = NE
- 68-112 = E
- 113-157 = SE
- 158-202 = S
- 203-247 = SW
- 248-292 = W
- 293-337 = NW